The Euclidean algorithm is not optimal among uniform processes

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Abstract

Van den Dries and Moschovakis conjectured that the Euclidean algorithm is optimal, up to a constant factor, among the algorithms that compute the greatest common divisor from the remainder function. Their homomorphism method gives lower bounds for all uniform processes, but for this problem it has given only $\Omega(\log\log a)$. We show that every pair $a > b \geq 1$ has a remainder-only derivation of $\gcd(a, b)$ with $O(\log a/\log\log a)$ remainder calls and leading coefficient $6 + o(1)$: one stored Euclidean prefix lets three-call contractions shrink every modulus. Hence the intrinsic remainder-call complexity of the gcd and of coprimeness is $O(\log a/\log\log a)$, and the homomorphism method cannot reach $\Omega(\log a)$. This refutes Conjecture 3.4 of Moschovakis's 2012 notes and the version (4F-5) of the Main Conjecture in his book. In the other direction, $\Omega(\log\log a)$ holds for almost all coprime pairs. For each $\varepsilon > 0$ we also give a fixed finite nondeterministic program with remainder cost at most $\varepsilon \log a + B_\varepsilon$ on all inputs $a \geq b \geq 1$; the program depends on ε. The Main Conjecture for algorithms remains open. The results are formalized in Lean 4; the cited general theory of uniform processes is not.

1 Introduction

Van den Dries and Moschovakis asked whether the Euclidean algorithm is optimal among its peers, the algorithms that compute the gcd from the remainder function [3]. Moschovakis's Main Conjecture asserts that for every algorithm α from $\mathrm{rem}, \mathrm{eq}_0$ computing $\gcd(x, y)$ there is an $r > 0$ such that α makes more than $r \log x$ remainder calls on infinitely many pairs $x > y \geq 1$ [5, p. 2]; here eq_0 is the zero test. Throughout, "the book" means Moschovakis's monograph [5], and logarithms are to base 2.

Derivations. A *derivation* from a, b is a finite list that starts with a, b and in which each further entry is $\mathrm{rem}(x, y)$ for earlier entries x, y with $y > 0$. Each further entry is one *instruction*, that is, one remainder call. Entries already in the list are *available* and may be reused; no constants and no other operations are supplied. Let $G(a, b)$ be the least number of instructions in a derivation from a, b that contains $\gcd(a, b)$, and let $D(a, b) = G(a, b)$ for coprime inputs. Thus $G(a, 0) = 0$ and $D(a, 1) = 0$. Arithmetic used in the proofs, such as subtraction, sorting or comparison, only selects instructions; it is never itself an instruction.

Theorem 1. *Uniformly for integers $a > b \geq 1$,*

$$G(a, b) = O\left(\frac{\log a}{\log\log a}\right).$$

In particular this bound holds for $D(a, b)$ when $\gcd(a, b) = 1$.

As usual, the asymptotic assertion concerns sufficiently large a; a bound valid at every positive input is $O(1+\log(2+a)/\log\log(4+a))$. By Proposition 8, the intrinsic remainder-call complexity of coprimeness equals D on positive coprime pairs, and that of the gcd lies between G and $G+2$. Hence Theorem 1 refutes Conjecture 3.4 of [2] and version (4F-5) of the Main Conjecture (Corollary 2).

Structures and intrinsic complexity. Write tt and ff for the truth values, and $\mathrm{eq}_c(x) = \mathrm{tt}$ if $x = c$ and $\mathrm{eq}_c(x) = \mathrm{ff}$ otherwise. We use the book's partial remainder operation: its second argument must be nonzero. Write $\mathcal{N}_0 = (\mathbb{N}, \mathrm{rem}, \mathrm{eq}_0)$ and $\mathcal{N}_\varepsilon = (\mathbb{N}, \mathrm{rem}, \mathrm{eq}_0, \mathrm{eq}_1)$; compare [5, (1C-10), pp. 24, 142–143]. The structure $\mathcal{N}_0$ is not the book's $\mathbf{N}_0$, whose expansion $\mathbf{N}_0[\div]$ appears below. Coprimeness is the relation $x \perp\!\!\!\perp y \iff x, y \geq 1 \wedge \gcd(x, y) = 1$ [5, §1A], [2]. Informally, a *certificate* for f at $\vec{x}$ in a structure $\mathcal{A}$ is a finite partial substructure of $\mathcal{A}$ generated by $\vec{x}$ on which every homomorphism into $\mathcal{A}$ respects f at $\vec{x}$; the definitions of certificates and intrinsic complexities are in [5, §§1D, 4C–4E] and are used in Section 5. Write $\mathrm{calls}_{\mathrm{rem}}(\mathcal{A}, f, \vec{x})$ for the least number of remainder facts in a certificate for f at $\vec{x}$ in $\mathcal{A}$. Uniform processes are the book's abstract model of algorithms from given primitives; every deterministic or nondeterministic recursive program induces one [5].

Background. The best known lower bound, $\Omega(\log\log a)$, is a logarithm below the conjectured $\Omega(\log a)$. For a recursive program deciding coprimeness from $=$, $<$, $+$, truncated subtraction, iq and rem, the depth of nested calls to these primitives exceeds $\frac{1}{10}\log\log a$ on infinitely many coprime pairs [3, Theorem A]; see also [4]. The book proves this as an intrinsic bound in $\mathbf{N}_0[\div] = (\mathbb{N}, 0, 1, =, <, +, \dot{-}, \mathrm{iq}, \mathrm{rem})$: $\mathrm{depth}_{\perp\!\!\!\perp}(\mathbf{N}_0[\div], a, b) > \frac{1}{10}\log\log a$ for every "difficult pair", a class that contains every solution of Pell's equation [5, Theorem 6C.5, p. 187]. Being intrinsic, the bound holds for all uniform processes, nondeterministic ones included.

A certificate in $\mathcal{N}_\varepsilon$ becomes one in $\mathbf{N}_0[\div]$ when each eq_0 or eq_1 fact is replaced by an equality with the constant 0 or 1, and its depth then exceeds its number of remainder facts by at most a constant. Hence $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b) = \Omega(\log\log a)$ on all difficult pairs. Pratt's nondeterministic algorithm [5, Theorem 2E.2 and Problems x2E.17–x2E.19] matches this $\Omega(\log\log a)$ bound up to a constant factor on successive Fibonacci numbers [5, Corollary 6C.7]. By contrast, for the binary gcd algorithm the same method gives a matching logarithmic lower bound: a recursive program deciding coprimeness from $=$, $<$, $+$, truncated subtraction, $2x$, $\mathrm{iq}(x, 2)$ and $\mathrm{rem}(x, 2)$ needs more than $\frac{1}{10}\log(a^2-1)$ calls on the pair (a, a^2-1), for every $a > 2$ [3, Theorem B]. Whether a logarithmic bound holds for all uniform processes was left open [5, p. 188], [2, Conjecture 3.4].

In a different model, Bshouty showed that a random access machine with linear operations, shifts and indirect addressing computes the gcd of two integers in $[0, N]$ in $O(\log N/\log\log N)$ operations, by reading precomputed blocks of the binary gcd algorithm from a table. He also proved that this is optimal: even with rational constants, multiplication and division by constants, and the floor function added, $\Omega(\log N/\log\log N)$ operations are necessary [1, Theorems i and ii]. The construction here uses no operation other than remainder and no table. Its saving of a factor $\log\log a$ comes instead from reusing one stored Euclidean prefix for every modulus.

The conjectures refuted. Moschovakis's 2012 notes state:

> **Conjecture 3.4** [2]. There is a rational number $r > 0$ such that for infinitely many $a \geq b \geq 1$, $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b) \geq r \log a$.

The book formulates one precise version of the Main Conjecture [5, p. 151]: there are a set $\mathcal{D} \subseteq \mathbb{N}^2$ and a $K > 0$ such that

$$(x, y) \in \mathcal{D} \implies \mathrm{calls}_{\mathrm{rem}}(\varepsilon, x, y) \leq L(x, y) \leq K \,\mathrm{calls}_{\mathrm{rem}}(\mathcal{A}, \gcd, x, y), \tag{4F-5}$$

where $L(x, y) = 2 \log \max(x, y)$, $\mathcal{A}$ is the structure ($\mathcal{N}_0$ or $\mathcal{N}_\varepsilon$), and ε denotes, as in the book, the Euclidean algorithm, and $\mathrm{calls}_{\mathrm{rem}}(\varepsilon, x, y)$ is the number of remainder calls it makes on (x, y). Elsewhere, apart from the subscript of $\mathcal{N}_\varepsilon$, ε denotes a small positive real. The book adds that $\mathcal{D}$ then satisfies its conditions (4F-3) and (4F-4). Condition (4F-4) makes the intrinsic complexity unbounded on $\mathcal{D}$, so the intended $\mathcal{D}$ is infinite; for finite $\mathcal{D}$ the statement would be trivial. Dropping the first inequality and asking only for an infinite $\mathcal{D}$ gives a weaker statement, and we refute that weaker statement.

Corollary 2. *The intrinsic complexities*

$$\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b), \qquad \mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_0, \gcd, a, b), \qquad \mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \gcd, a, b)$$

are $O(\log a / \log\log a)$, uniformly for $a \geq b \geq 1$. Consequently Conjecture 3.4 of [2] is false, and so is the version (4F-5) of the Main Conjecture with $\mathcal{A} = \mathcal{N}_0$ or $\mathcal{A} = \mathcal{N}_\varepsilon$: for every $r > 0$ only finitely many pairs $a \geq b \geq 1$ satisfy $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b) \geq r \log a$, and for every $K > 0$ only finitely many pairs $(x, y) \in \mathbb{N}^2$ satisfy $L(x, y) \leq K \,\mathrm{calls}_{\mathrm{rem}}(\mathcal{A}, \gcd, x, y)$.

The corollary follows from Theorem 1 and Proposition 8; for noncoprime pairs, the bound on the coprimeness complexity comes from Remark 9. For (4F-5) it covers all pairs in $\mathbb{N}^2$: the gcd is symmetric, and a pair with a zero entry is certified by the single fact $\mathrm{eq}_0(0) = \mathrm{tt}$, with no remainder facts. The corollary is formalized, and so are refutations of the literal statements: Conjecture 3.4 with r rational and with infinitely many values of a (not merely of pairs), and (4F-5) for $\mathcal{N}_\varepsilon$ with a set $\mathcal{D}$ on which the intrinsic complexity is unbounded.

Theorem 4F.1 [5, pp. 149–150] supplies a single uniform process that attains these pointwise intrinsic bounds. Hence some uniform processes of $\mathcal{N}_\varepsilon$ decide coprimeness and compute the gcd with $O(\log a / \log\log a)$ remainder calls on all pairs $a \geq b \geq 1$. This settles the possibility raised after Corollary 6C.7 [5, p. 188]: the Main Conjecture does not hold for all uniform processes of $\mathcal{N}_\varepsilon$.

The limit of the homomorphism method. The homomorphism method proves lower bounds on intrinsic complexities. These bound every uniform process that computes the function: for any of the book's complexity measures μ, such as $\mathrm{calls}_{\mathrm{rem}}$, if α computes f in $\mathcal{A}$ with cost $\mu(\alpha, \vec{x})$ on $\vec{x}$, then $\mu(\mathcal{A}, f, \vec{x}) \leq \mu(\alpha, \vec{x})$, in particular for every deterministic or nondeterministic recursive program [5, Proposition 4E.1, Corollary 4E.2]. The Homomorphism Test [5, Lemma 4E.3] proves a lower bound on $\mu(\mathcal{A}, f, \vec{x})$ itself, by showing that every generated diagram below the proposed cost admits a homomorphism that fails to preserve the answer. Corollary 2 caps every such argument.

Corollary 3 (Limit of the homomorphism method)**.** *Let $\mathcal{A} = \mathcal{N}_0$ or $\mathcal{A} = \mathcal{N}_\varepsilon$, and let $g(a, b)$ be any lower bound for $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b)$ or for $\mathrm{calls}_{\mathrm{rem}}(\mathcal{A}, \gcd, a, b)$. Then $g(a, b) = O(\log a / \log\log a)$, uniformly for $a \geq b \geq 1$.*

Proof. Immediate from Corollary 2. □

Remark 4. By Corollary 3, no lower bound on intrinsic complexity shows that every algorithm from $\mathrm{rem}, \mathrm{eq}_0$ computing the gcd makes at least $r \log a$ remainder calls on infinitely many pairs. A proof of the Main Conjecture must use properties of algorithms that uniform processes need not have, such as being induced by a finite program, or determinism.

Fixed programs. There is also an algorithmic consequence with a different quantifier order. Section 7 constructs fixed finite nondeterministic recursive programs with arbitrarily small logarithmic remainder-call coefficients, uniformly over all inputs. The programs use a fixed-length prefix and ordinary recursive calls. Pratt's algorithm is known to attain $O(\log\log a)$ on successive Fibonacci numbers; here the logarithmic coefficient holds on all inputs and is made small by enlarging the program.

Idea of the proof. Let $a > b$ be coprime with $a \le N = 2^n$, and store the Euclidean remainders $r_0 = a, r_1 = b, \dots, r_h$, where $h = \lceil \sqrt{n} \rceil$. If one of them is at most the early threshold $N^{3/4}$, recurse on that consecutive pair. Otherwise the consecutive pairs of the prefix stay distinct modulo every $m > N^{1/3}$ (Lemma 6), so the residues $r_i \bmod m$ take at least $\sqrt{h}$ values, and three remainder calls produce a new positive modulus below $m/(\sqrt{h}-1)$ (Lemma 5). Repeating this from $m = r_h$ reaches the target modulus $N^{1/3}$ within $O(n/\log n)$ calls, and two recursive gcd computations on inputs at most $N^{1/3}$ finish. The resulting recurrence (5) has solution $O(n/\log n) = O(\log a/\log\log a)$.

agraphOrganization. Section 2 reduces to coprime inputs. Section 3 proves the two arithmetic lemmas, and Section 4 combines them into a recurrence that proves Theorem 1 and the leading coefficient $6 + o(1)$. Section 5 relates derivations to certificates. Section 6 proves the almost-all lower bound, and Section 7 constructs the fixed programs. Section 8 discusses open problems, and Section 9 reports computational checks. Appendix A maps the numbered results to their Lean 4 formalizations.

2 Normalization

Write $g = \gcd(a, b) > 0$. For positive y,

$$\operatorname{rem}(gx, gy) = g \operatorname{rem}(x, y).$$

Every generated value is divisible by g. Consequently scaling a derivation, or dividing all its values by g, preserves its length and the validity of every instruction. Hence

$$G(a, b) = D(a/g, b/g). \tag{1}$$

Define

$$T(n) = \max_{0 \le b \le a \le 2^n,\ a > 0} G(a, b).$$

Each $G(a, b)$ is defined, since the Euclidean remainders form a derivation that contains $\gcd(a, b)$, so $T(n)$ is a finite maximum. By (1) it suffices to bound D; the cases $a = b$, $b = 0$ and $b = 1$ need no instructions.

3 Three-call contractions

Lemma 5. *If $m > 0$ is available and a finite collection S of available values occupies $t \ge 2$ residue classes modulo m, at most three instructions generate w with*

$$0 < w < \frac{m}{t-1}.$$

Proof. Sort the distinct residues. Some adjacent pair $u < v$ has gap $\delta = v - u < m/(t-1)$, because the sum of the $t-1$ adjacent gaps is less than m. Each selected residue is $\operatorname{rem}(x, m)$ for some $x \in S$, so it costs one instruction. If $u = 0$, use v. If $0 < u \le \delta$, use u. Otherwise $0 < \delta < u$ and $v = u + \delta$, so $\operatorname{rem}(v, u) = \delta$. This uses at most three instructions. □

Lemma 6 (Euclidean-prefix lemma)**.** *Let $r_0 = a > r_1 = b > \cdots > r_h > 0$ be a Euclidean prefix of a coprime pair, with $r_{i-1} = q_i r_i + r_{i+1}$ and $q_i \geq 1$ for $1 \leq i < h$. For every integer $m > 2a/r_h$, the h pairs*

$$(r_i \bmod m, r_{i+1} \bmod m), \qquad 0 \leq i < h,$$

are distinct. The prefix therefore occupies t residue classes with $t^2 \geq h$.

Proof. Suppose the pairs at $i < j$ coincide. Set

$$M = \prod_{\ell=i+1}^{j} \begin{pmatrix} q_\ell & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} A & B \\ C & E \end{pmatrix}, \qquad \begin{pmatrix} r_i \\ r_{i+1} \end{pmatrix} = M \begin{pmatrix} r_j \\ r_{j+1} \end{pmatrix}.$$

Right multiplication by $\left(\begin{smallmatrix} q & 1 \\ 1 & 0 \end{smallmatrix}\right)$ sends (A, B, C, E) to $(qA+B,\ A,\ qC+E,\ C)$, starting from $(q, 1, 1, 0)$ for a single factor. By induction, all entries are nonnegative, $A \geq B \geq E$, $A \geq C \geq 1$, $B \geq 1$, and $\det M = (-1)^{j-i}$. Also $Ar_j \leq r_i \leq a$ and $r_j > r_h$ (as $j < h$), whence $A < a/r_h$. Write

$$\Delta = \det(M - I) = 1 + \det M - A - E.$$

For an odd number of factors, $\Delta = -A - E < 0$. By the same recursion, a product of at least two factors has $A \geq 2$ and $E \geq 1$; thus for an even positive number of factors, $\Delta = 2 - A - E < 0$. In either case,

$$0 < |\Delta| \leq A + E \leq 2A \leq 2a/r_h < m.$$

The presumed congruence gives $(M - I)(r_j, r_{j+1})^{\mathsf{T}} \equiv 0 \pmod m$. Multiplication by the adjugate shows that m divides both Δr_j and Δr_{j+1}. Consecutive remainders are coprime, so Bézout's identity implies $m \mid \Delta$, a contradiction. Finally, t individual residue classes give at most t^2 ordered pairs. □

Thus a stored prefix can be reused for every available modulus $m > 2a/r_h$. When $\sqrt{h} > 1$, Lemma 5 applied to the prefix gives three instructions that generate

$$0 < w < \frac{m}{\sqrt{h} - 1}. \tag{2}$$

We call these three instructions a *contraction* of m, and w the new modulus. The universal quantifier over m is essential: the modulus generated by a contraction is not predetermined. In Lemma 6 the prefix length h is generic; Section 4 chooses $h = \lceil \sqrt{n} \rceil$, while Corollary 7 and Section 7 choose $h = k^2 + 1$.

4 Construction and recurrence

Fix an integer $n > 16$ and put $N = 2^n$ and $h = \lceil \sqrt{n} \rceil$, so that $h > 4$; this h is the prefix length to which Lemma 6 will be applied. After normalization we may assume that $a > b > 1$ are coprime and $a \leq N$. The derivation has three phases.

(i) Store a Euclidean prefix of length at most h. If it reaches a positive remainder at most the *early threshold* $N^{3/4}$, recurse on that smaller consecutive pair; this gives (3).

(ii) Otherwise contract the modulus with (2), starting at $m = r_h$, until $0 < m \leq N^{1/3}$, the *target modulus*.

(iii) Finish with two recursive gcd derivations on inputs at most m; phases (ii) and (iii) together give (4).

Phase (i): the prefix. Generate the Euclidean remainders $r_2, r_3, \ldots$, one instruction each, and stop as soon as the most recent value is positive and at most $N^{3/4}$, or once r_h is available. The test is also applied to the input $r_1 = b$, so if $b \le N^{3/4}$ no instruction is used. This preliminary work costs at most h instructions. Since the inputs are coprime, Euclid's algorithm reaches the positive remainder 1 before it can reach zero, so the rule stops before a zero remainder appears.

If a positive remainder $c \le N^{3/4}$ is encountered, generate the next remainder $s < c$ with one more instruction. The pair (c, s) is coprime. If $s = 0$, then $c = 1$ is already available; otherwise both inputs to the remaining derivation are at most $2^{\lceil 3n/4 \rceil}$. This gives the bound

$$h + 1 + T(\lceil 3n/4 \rceil). \tag{3}$$

Phase (ii): contractions. Otherwise $r_h > N^{3/4}$, so $2a/r_h < 2N^{1/4}$. For $n > 12$, $N^{1/3} > 2N^{1/4}$, and Lemma 6 applies to every available $m > N^{1/3}$. Starting at $m = r_h$, repeatedly replace m by the value w of (2) until $0 < m \le N^{1/3}$. Set

$$\kappa_n = \sqrt{h} - 1 > 1, \qquad J_n = \left\lceil \frac{2n/3}{\log \kappa_n} \right\rceil.$$

There are at most J_n contractions: if J_n have occurred then $m < N/\kappa_n^{J_n} \le N^{1/3}$. Since $\log \kappa_n = \frac{1}{4}\log n + O(1)$, their cost is $3J_n = O(n/\log n)$.

Phase (iii): two recursive derivations. The small m need not be coprime to either input. Generate $u = \mathrm{rem}(a, m)$ and $v = \mathrm{rem}(b, m)$, then concatenate derivations of $d = \gcd(m, u)$ and $e = \gcd(d, v)$; no gcd-oracle instruction is used. Since $m \le 2^{\lceil n/3 \rceil}$ and G is symmetric, each has a remainder-only derivation of length at most $T(\lceil n/3 \rceil)$. Each sub-derivation starts from values already in the list (m, u, and then d, v), so it can be appended unchanged. The gcd identities give

$$e = \gcd(\gcd(m, a \bmod m), b \bmod m) = \gcd(m, \gcd(a, b)) = 1.$$

The cost in this case is at most

$$h + 3J_n + 2 + 2T(\lceil n/3 \rceil). \tag{4}$$

The recurrence. The two case estimates (3) and (4) and normalization imply

$$T(n) \le \max\{T(\lceil 3n/4 \rceil) + O(\sqrt{n}),\ 2T(\lceil n/3 \rceil) + O(n/\log n)\}. \tag{5}$$

All implied constants are independent of the inputs.

Proof of Theorem 1. For $n \ge 2$ let $f(n) = n/\log n$. The ratios

$$\frac{f(\lceil 3n/4 \rceil)}{f(n)} \longrightarrow \frac{3}{4}, \qquad \frac{2f(\lceil n/3 \rceil)}{f(n)} \longrightarrow \frac{2}{3}$$

show that, beyond a fixed threshold, $f(\lceil 3n/4 \rceil) \le \frac{7}{8} f(n)$ and $2f(\lceil n/3 \rceil) \le \frac{5}{6} f(n)$. Both overheads in (5) are at most $c_0 f(n)$ for a fixed c_0. Choose the induction threshold above 16, where the recurrence holds, and large enough that both recursive indices are at least 2 and strictly less than n. Choose $C \ge 8c_0$, increased to cover the finitely many values below the threshold. Strong induction then gives $T(n) \le Cf(n)$: the two bounds are $(7C/8 + c_0)f(n)$ and $(5C/6 + c_0)f(n)$, both at most $Cf(n)$. Finally take $n = \lceil \log a \rceil$. This also gives $G(a, b) = o(\log a)$ uniformly as $a \to \infty$. □

4.1 The leading coefficient

The same construction allows a more precise bound without changing either arithmetic lemma.

Corollary 7. *For every $\varepsilon > 0$, there is an a_0 such that for every $a \geq a_0$ and $0 \leq b \leq a$,*

$$G(a,b) \leq (6+\varepsilon)\frac{\log a}{\log\log a}.$$

Equivalently, the uniform upper bound has coefficient $6 + o(1)$.

Proof. Fix $0 < c < 1/2$ and $0 < \beta < 1/2$, and put $\alpha = 1 - \beta/2$. For sufficiently large n, choose the prefix length h afresh, together with auxiliary parameters, as

$$p = \lfloor c \log n \rfloor, \qquad k = 2^p, \qquad h = k^2 + 1, \qquad J = \lceil n/p \rceil.$$

Use N^α as the early threshold and N^β as the target modulus, where $N = 2^n$. In the late case $r_h > N^\alpha$, so $2a/r_h < 2N^{1-\alpha} = 2N^{\beta/2} < N^\beta$ for sufficiently large n, and the prefix lemma applies to every modulus above N^β. Since $k^2 < h$, the prefix occupies at least $k+1$ residue classes, so each contraction reduces the modulus by a factor greater than k. Moreover $k^J \geq N$, so J contractions suffice. Writing $f(n) = n/\log n$, we have

$$h = O(n^{2c}) = o(f(n)), \qquad 3J = \left(\frac{3}{c} + o(1)\right) f(n).$$

The two branches therefore give

$$T(n) \leq \max\left\{ T(\lceil \alpha n \rceil) + o(f(n)), \quad 2T(\lceil \beta n \rceil) + \left(\frac{3}{c} + o(1)\right) f(n) \right\}.$$

Theorem 1 implies that $L = \limsup_{n\to\infty} T(n)/f(n)$ is finite and nonnegative. For each fixed $q > 0$, the ratio $f(\lceil qn \rceil)/f(n)$ tends to q. Taking upper limits in the recurrence yields

$$L \leq \max\{\alpha L,\ 2\beta L + 3/c\}.$$

If the first term is the larger, $L \leq \alpha L$ forces $L = 0$, as $\alpha < 1$; otherwise $L(1-2\beta) \leq 3/c$. In either case $L(1-2\beta) \leq 3/c$. First let $\beta \downarrow 0$ and then $c \uparrow 1/2$ to obtain $L \leq 6$. Finally, $n = \lceil \log a \rceil$ satisfies $f(n) \sim \log a/\log\log a$, proving the assertion. All upper limits concern the worst-case function T, so the resulting threshold is independent of b. □

No optimality of the coefficient 6 is asserted. The parameters of Section 4 suffice for the simpler Big-O proof above.

5 Certificates and uniform processes

This section shows that certificates for coprimeness correspond to derivations of 1, and certificates for the gcd to derivations of the gcd with two extra remainder facts. A derivation determines a finite generated partial substructure by recording its remainder equations. Repeated equations can be discarded.

Proposition 8. *For coprime $a, b \geq 1$, $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b) = D(a,b)$. For $(a,b) \neq (0,0)$,*

$$G(a,b) \leq \mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_0, \gcd, a, b) \leq G(a,b) + 2,$$

and the same bounds hold for $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \gcd, a, b)$.

Proof. Write $\gcd{=}1$ for the relation $\gcd(x,y)=1$ on all of $\mathbb{N}^2$ (unlike $\perp\!\!\!\perp$, it holds at $(1,0)$). A derivation of 1 together with $\mathrm{eq}_1(1)=\mathrm{tt}$ certifies the relation: every common divisor of the images of the inputs divides every transported generated value, hence divides 1. Conversely, without a positive eq_1 fact the map $x\mapsto 2x$ preserves every recorded remainder and zero-test fact, as well as every negative one-test fact. It cannot preserve coprimeness. A certificate therefore contains a positive eq_1 fact, hence the value 1. Enumerating a finite generated diagram in generation order gives a derivation using no more calls than the number of remainder facts. Consequently, on coprime inputs, writing $\mathrm{calls}(\cdot)$ for the number of all primitive facts of a certificate, not only remainder facts,

$$\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon,\gcd{=}1,a,b)=D(a,b),\qquad \mathrm{calls}(\mathcal{N}_\varepsilon,\gcd{=}1,a,b)=D(a,b)+1.$$

For the book's relation $\perp\!\!\!\perp$, which also requires $a,b\geq 1$, it suffices to add the facts $\mathrm{eq}_0(a)=\mathrm{ff}$ and $\mathrm{eq}_0(b)=\mathrm{ff}$. Conversely, when $a\perp\!\!\!\perp b$, every certificate for $\perp\!\!\!\perp$ also certifies $\gcd(a,b)=1$. So for coprime $a,b\geq 1$ the remainder count is unchanged, $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon,\perp\!\!\!\perp,a,b)=D(a,b)$, and the total count is at most $D(a,b)+3$.

For gcd as a function, the certificate must reach its output g, giving a lower bound of $G(a,b)$. For an upper bound, append

$$\mathrm{rem}(a,g)=z,\qquad \mathrm{rem}(b,g)=z,\qquad \mathrm{eq}_0(z)=\mathrm{tt},$$

where the generated z is zero. Every homomorphism π has $\pi(z)=0$ and $\pi(g)>0$ (by definedness of either appended remainder). Thus $\pi(g)$ divides both input images. Conversely every common divisor of those images divides $\pi(g)$ by induction over the derivation. Therefore, for $(a,b)\neq(0,0)$,

$$G(a,b)\leq \mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_0,\gcd,a,b)\leq G(a,b)+2.$$

The lower-bound argument applies to every gcd certificate, eq_1 facts included, and allowing eq_1 facts can only lower the count, so the upper bound transfers to $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon,\gcd,a,b)$. □

Remark 9. At $(0,0)$ the output is an input and the empty certificate suffices. If the gcd is taken as a function on positive inputs only, the facts $\mathrm{eq}_0(a)=\mathrm{ff}$ and $\mathrm{eq}_0(b)=\mathrm{ff}$ may be appended. For a noncoprime input, adding $\mathrm{eq}_1(g)=\mathrm{ff}$ to the gcd certificate certifies noncoprimeness, so the sublogarithmic remainder-call bound holds for both truth values of coprimeness as well as for the gcd. The overhead is a fixed additive constant and does not change the leading coefficient, so Corollary 7 holds for all three complexities of Corollary 2 on positive inputs.

6 An almost-all lower bound

The difficult pairs of [5, Theorem 6C.5] give $\Omega(\log\log a)$ on a sparse set. An elementary Diophantine argument gives the same order for almost all coprime pairs, simultaneously for every derivation.

Proposition 10. *For every $d\geq 1$ and $N\geq 3$, at most $16(2+\ln N)N^{2-2^{-d}}$ coprime pairs $2\leq b<a\leq N$ satisfy $D(a,b)\leq d$. Hence $D(a,b)\geq(1-o(1))\log\log N$ for all but $o(N^2)$ coprime pairs below N.*

Proof. For $B\geq 1$ call (a,b) *B-regular* if $|qb-pa|\geq a/(Bq)$ whenever $q\geq 1$, $p\in\mathbb{Z}$ and $qb\neq pa$.

- **Small values have large coefficients.** Every generated value is an integer form $v=ua+wb$. If (a,b) is B-regular and $v>0$, then $v\geq a/(B(|u|+|w|))$: for $w\neq 0$ apply regularity with $q=|w|$, and for $w=0$ note $v\geq a$.

- **Coefficients grow at most doubly exponentially.** Let H_t bound $|u|+|w|$ over the values available after t instructions, so $H_0 = 1$. An instruction $r = \mathrm{rem}(x, y)$, with $x = sy + r$, has quotient $s \le a/y \le BH_t$, hence $H_{t+1} \le H_t + BH_t^2 \le 2BH_t^2$ and $H_t \le (2B)^{2^t-1}$.
- **Regular pairs need long derivations.** Producing 1 within d instructions forces $1 \ge a/(BH_d)$, that is $a \le (2B)^{2^d}/2$.
- **Few pairs are irregular.** Suppose $0 < |qb - pa| < a/(Bq)$. Dividing p, q by their gcd preserves this, and it forces $q < a/B \le N/B$ and $0 \le p \le q$. For each q there are at most $2q$ values of p and $2N/(Bq)$ values of $r = qb - pa$. Given p, q, r with $\gcd(p, q) = 1$, a lies in one residue class modulo q, which leaves at most $N/q + 1$ choices, and b is then determined. So at most $(4N^2/B)(2 + \ln N)$ pairs below N are irregular.
- **Dyadic ranges.** For $X/2 < a \le X$ take $B = X^{2^{-d}}/2$. If $B \ge 1$, a regular pair with $D \le d$ would have $a \le X/2$, so every pair in the range with $D \le d$ is irregular. That gives at most $8(2 + \ln X)X^{2-2^{-d}}$ pairs, and the same bound is trivial when $B < 1$. Summing over $X = N, N/2, \ldots$ gives the first claim.

For the second claim take $d = \lfloor \log_2 \ln N - 2 \log_2 \ln \ln N \rfloor$. Then $N^{-2^{-d}} \le \exp(-(\ln \ln N)^2)$, so the bound is $o(N^2)$. □

Proposition 10 is formalized, with a slightly larger constant; see Appendix A. The saving 2^{-d} in the exponent of $N^{2-2^{-d}}$ shrinks as d grows, so this argument cannot reach the scale $\log N / \log \log N$ of Theorem 1, let alone $\log N$.

7 Fixed programs with arbitrarily small coefficients

For a nondeterministic program P, let $c_P(a, b)$ be the minimum number of remainder calls among its convergent computations. We require at least one such computation on each input and require every convergent computation to return the gcd. These are the conventions of [5, (2E-2), (2E-5)].

Theorem 11. *For each integer $p \ge 1$, there is a finite nondeterministic recursive program P_p over $(\mathbb{N}, \mathrm{rem}, \mathrm{eq}_0)$ computing the gcd such that*

$$\limsup_{a \to \infty} \max_{1 \le b \le a} \frac{c_{P_p}(a, b)}{\log a} \le \frac{3}{p}.$$

Its generator, defined below, has $W = 4^p + 5$ individual parameters, and no internal function has larger arity. Consequently, for every $\varepsilon > 0$ there are a fixed program P and a constant $B \ge 0$ such that

$$\forall a \ge b \ge 1, \qquad c_P(a, b) \le \varepsilon \log a + B.$$

The program. Fix $W \ge 3$ and write $\vec{x} = (x_0, \ldots, x_{W-1})$ and $\vec{x}[i \leftarrow E]$ for the tuple with its ith expression replaced by E. The generator is given by the finite alternatives

$$\begin{aligned}
\mathrm{choose}(\vec{x}) &= x_i && (0 \le i < W), && \text{(C)}\\
\mathrm{gen}(\vec{x}) &= \mathrm{choose}(\vec{x}), && && \text{(G1)}\\
\mathrm{gen}(\vec{x}) &= \mathrm{gen}(\vec{x}[i \leftarrow \mathrm{rem}(\mathrm{choose}(\vec{x}), \mathrm{choose}(\vec{x}))]) && (0 \le i < W), && \text{(G2)}\\
\mathrm{gen}(\vec{x}) &= \mathrm{gen}(\vec{x}[i \leftarrow \mathrm{gen}(\vec{x})]) && (0 \le i < W), && \text{(G3)}\\
\mathrm{gen}(\vec{x}) &= \mathrm{gen}(\mathrm{choose}(\vec{x}), \ldots, \mathrm{choose}(\vec{x})). && && \text{(G4)}
\end{aligned}$$

There are W independent selector calls in (G4). Tuple notation and indices are syntax abbreviations: after fixing W, each line expands into ordinary first-order terms with fixed arities. No tuple, index, numeric constant, or update operation is supplied as runtime data or a primitive. The nested generator in (G3) is evaluated once; its caller's other parameters remain available when it returns. The recursion stack can grow, so W is an arity bound, not a total storage bound.

Add $\mathrm{loop}(z) = \mathrm{loop}(z)$ and define $\mathrm{accept}(a, b, g)$ by nested zero-test conditionals: if $g = 0$, loop; otherwise test $\mathrm{rem}(a, g) = 0$ and then $\mathrm{rem}(b, g) = 0$, returning g if both tests succeed and looping otherwise. The head is

$$P_W(a, b) = \text{if } \mathrm{eq}_0(b) \text{ then } a \text{ else } \mathrm{accept}(a, b, \mathrm{gen}(a, b, a, \ldots, a)).$$

The four internal symbols have arities $W, W, 3, 1$ (for choose, gen, accept, loop). Every common divisor of the generator's parameters divides its output, by induction on a finite evaluation. Thus a candidate is divisible by $\gcd(a, b)$; successful verification proves the reverse divisibility. Every convergent computation therefore returns the gcd. There is always a convergent computation following Euclid, using parameter selection and copying to retain the current pair. For $b > 0$, final verification costs exactly two remainder calls.

This syntax is a finite recursive program in the sense of [5, §§2A, 2E]; Appendix A records the Lean check that its evaluation trees correspond, with equal remainder cost, to convergent computations of a specialization of the book's recursive machine.

Proof of Theorem 11. We follow the construction of Section 4: store the prefix, contract, then recurse twice through nested generator calls. Take $P_p = P_W$ with $W = 4^p + 5$; that is, set $k = 2^p$, $h = k^2+1$ (the prefix length, as in Corollary 7 but now with p fixed), and $W = h+4$. Store $r_0, \ldots, r_h$ in $h+1$ parameters, reserving three parameters for the modulus and two scratch residues. Lemmas 5 and 6 give a contraction by a factor greater than k in at most three remainder calls, preserving the prefix. The program can choose precisely the operands of each contraction.

The *generator cost* of (a, b) is the minimum number of remainder calls in an evaluation of gen that returns $\gcd(a, b)$ from $(a, b, a, \ldots, a)$. Let $T_W(n)$ be its maximum over $a, b \le 2^n$. Euclid gives $T_W(n) = O(n)$. To *route* to x, y means to apply (G4) to pass to the tuple $(x, y, x, \ldots, x)$, at no remainder cost. Copying parameters gives symmetry. Multiplying every value by $g = \gcd(a, b)$ turns a run on $(a/g, b/g)$ that returns 1 into a run on (a, b) that returns g, with the same remainder count. It therefore suffices to analyze coprime inputs.

Fix $0 < \beta < 1/2$ and put $\alpha = 1 - \beta/2$. Use the early threshold N^α and the target modulus N^β, where $N = 2^n$. If no prefix remainder is at most N^α, then $r_h > N^\alpha$ and $2a/r_h < 2N^{1-\alpha} < N^\beta$ for all sufficiently large n. There are at least $k+1$ distinct prefix residues modulo every modulus used before stopping, since $h > k^2$. At most $\lceil n/p \rceil$ contractions produce a positive $m \le N^\beta$.

In the early case, route directly to the smaller consecutive Euclidean pair. Otherwise form $u = a \bmod m$ and $v = b \bmod m$ in the scratch positions. A nested generator call, with its parameters routed to copies of m, u, returns $d = \gcd(m, u)$ while its caller retains v. Replace the modulus parameter by d and route to copies of d, v for the second gcd computation. The result is $\gcd(m, a, b) = 1$. All these routes and nested calls are available in the same fixed program. Thus, eventually in n,

$$T_W(n) \le \max\left\{T_W(\lceil \alpha n \rceil) + h + 1, \quad 2T_W(\lceil \beta n \rceil) + h + 3\lceil n/p \rceil + 2\right\}.$$

Set $L_W = \limsup_{n\to\infty} T_W(n)/n$, finite and nonnegative by Euclid. Since h is fixed, division by n and passage to upper limits gives

$$L_W \le \max\{\alpha L_W,\ 2\beta L_W + 3/p\}.$$

If the first term is the larger, $L_W \le \alpha L_W$ forces $L_W = 0$, as $\alpha < 1$; otherwise $(1 - 2\beta)L_W \le 3/p$. The program is independent of β, so letting $\beta \downarrow 0$ gives $L_W \le 3/p$. Take $n = \lceil \log a \rceil$ and add the two verification calls. Finally, choose $p > 3/\varepsilon$ and absorb the finitely many smaller inputs in B. □

8 Discussion and open problems

The gap. With the lower bound recalled in Section 1, the intrinsic complexity of coprimeness in $\mathcal{N}_\varepsilon$ now lies between $\Omega(\log \log a)$, on infinitely many pairs and in fact on almost all coprime pairs (Proposition 10), and $O(\log a / \log \log a)$, on all pairs. Closing this gap is an open problem. In particular, it is not known whether the bound of Theorem 1 is optimal for remainder-only derivations, as the corresponding bound is in Bshouty's model.

Beyond the homomorphism method. Corollary 3 does not exclude arguments that combine homomorphisms with additional restrictions on computations. Corollaries 2 and 3 also do not settle the form of the Main Conjecture for deterministic uniform processes [2, Conjecture 3.9], [5, §4H and p. 188].

Uncharged choice. Selecting a close pair of residues is uncharged existential choice in a straight-line derivation. A deterministic remainder-only algorithm cannot assume that choice is available at constant cost. Computing all prefix residues would take about h calls per contraction. The programs of Section 7 make the selection nondeterministically, but only with a prefix of fixed length. This leaves a further question: can one fixed first-order recursive program store and repeatedly access a prefix whose length grows with the input, and so attain $O(\log a / \log \log a)$ remainder calls? Proposition 2E.1 converts each certificate into a computation of some program; it does not supply one program for all certificates [5, §2E]. The book also distinguishes the best uniform process from a nondeterministic program [5, p. 150]. The construction therefore settles neither algorithmic version of the Main Conjecture: the Main Conjecture for algorithms, that is, recursive programs, deterministic or nondeterministic [5, p. 2], remains open.

Quantifier order. Theorem 11 is a statement about a family of programs: the prefix length grows when a smaller coefficient is requested, but remains fixed as the inputs grow, and no single sublogarithmic program is obtained. The Main Conjecture allows its lower-bound coefficient to depend on the program, so Theorem 11 is compatible with it. Neither the coefficient nor the arity bound is claimed optimal. Merely inlining several remainder instructions into one term would still charge for each external remainder call; the saving here uses the prefix lemma and recursive reduction.

9 Computational checks

The offline Python planner `remainder_only_certificate.py` in the accompanying repository uses the parameters of Section 4 and unrestricted arithmetic to choose instructions. An independent replay then checks operand availability, nonzero divisors, every remainder equation and the final gcd. Below a base size (256 bits by default) it uses Euclid's algorithm. On one pseudorandom coprime pair of each size, generated from its default seed, it gives these instruction counts:

Input bits	This construction	Euclid (to reach 1)
512	196	326
2048	669	1204
8192	2059	4754

Random pairs are not Euclid's worst case: on successive Fibonacci numbers Euclid needs about $1.44 \log a$ calls. These are checks of finite instances, not an asymptotic proof. A reproducible suite of 310 cases also passes, including 175 noncoprime pairs, with base size 64 bits and 3,543 contractions in total, and four deliberately invalid derivations are rejected. The Euclidean-prefix lemma was also checked exhaustively for every coprime pair with $3 \le a < 1500$ and $1 \le b < a$, every prefix length, and up to 60 consecutive moduli above the threshold $2a/r_h$, restricted to $m < 4a$ (254,722,179 instances, no failures).

For the programs of Section 7, an independent Python compiler and interpreter, which evaluates arguments right to left as the book's machine does, check 138 cases across three values of W, 129 additional scaled replays, and eight negative cases. The interpreter is not formally verified; unrestricted arithmetic used to plan successful choices is not part of the programs.

A Formal verification

The Lean development uses Lean 4.35.0 with mathlib at its v4.35.0 tag. Table 1 maps the numbered results of the paper to Lean names. Module names are given in full. Proposition 10 is formalized with the constant 40 in place of 16. The development contains no proof placeholders or added axioms. An axiom audit of the main results reports only the standard Lean foundations `propext`, `Classical.choice`, and `Quot.sound`. The general theory of uniform processes, Theorem 4F.1 and the lower bounds cited in Section 1 are cited, not formalized; so is the consequence of Theorem 4F.1 drawn after Corollary 2. Theorem 4F.1 is needed only to restate the intrinsic bounds in terms of a single best uniform process, not for the refutations themselves.

A.1 Bounds, certificates and refutations

The theorem `uniform_real_bound` states

$$\exists C > 0 \; \forall a, b \in \mathbb{N}, \quad 4 \le a \;\wedge\; b \le a \implies G(a,b) \le C \frac{\log_2 a}{\log_2(\log_2 a)}.$$

The sharper theorem `uniform_real_bound_six` states that for every $\varepsilon > 0$, eventually in a, the same inequality holds for all $b \le a$ with $C = 6 + \varepsilon$, and `uniform_intrinsic_bound_six` gives the corresponding statement for the three intrinsic complexities on positive inputs. The proof starts from a finite-store instruction semantics in which the only instruction is remainder with a positive, available divisor. Costs are lengths of explicit derivations, not values of an assumed recurrence. A formal conversion to finite generated diagrams proves the gcd certificate bound with two extra remainder facts and the bound for positive and negative coprimeness instances. The reduction itself is also checked: on coprime inputs, the least number of remainder facts in a certificate for $\gcd(a,b) = 1$ equals $D(a,b)$, and G agrees with D.

The Lean definitions from certificates are `callsCoprimeness` for the book's relation $\perp\!\!\!\perp$ in $\mathcal{N}_\varepsilon$, with positive inputs, and `callsGcd` and `callsRemGcd` for the gcd in $\mathcal{N}_0$ and $\mathcal{N}_\varepsilon$. Adding the facts $\mathrm{eq}_0(a) = \mathrm{ff}$ and $\mathrm{eq}_0(b) = \mathrm{ff}$ is checked to turn a certificate for $\gcd(a,b) = 1$ into one for $\perp\!\!\!\perp$, so on positive coprime pairs the complexity agrees with the reduction.

Table 1: Results of the paper and the corresponding Lean declarations. All names are in the namespace `ShortGCD`.

Paper	Lean name(s)
G and D (Section 1)	`G`, `D`
Theorem 1	`uniform_real_bound`; worst-case Big-O form `sizeCost_isBigO`, stating that $\max_{a,b\le M} G(a,b)$ is $O(\log M/\log\log M)$
Normalization (1)	`G_normalize`
Lemma 5	`small_remainder`
Lemma 6	`prefix_pairs_distinct`, `residue_card_sq`
Corollary 7	`uniform_real_bound_six`
$\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b)$	`callsCoprimeness`, with `BookCoprime`
$\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_0, \gcd, a, b)$, $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \gcd, a, b)$	`callsGcd`, `callsRemGcd`
$\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \gcd=1, a, b)$	`callsRem`
Reduction to D (Proposition 8)	`callsRem_eq_D`, `G_eq_D`, `certifiesCoprimeness_of_certifiesCoprime`, `callsCoprimeness_eq_callsRem`
Certificate bounds (Proposition 8)	`callsCoprimeness_le`, `callsGcd_le`, `callsRemGcd_le_callsGcd`; the lower bound for $\mathcal{N}_\varepsilon$ is `gcd_certificate_lower` applied to certificates with eq_1 facts
Corollary 2	`not_conjecture_3_4`, `not_main_conjecture_4F5` ($\mathcal{N}_0$), `not_main_conjecture_4F5_Neps` ($\mathcal{N}_\varepsilon$); literal readings `not_conjecture_3_4_literal`, `not_main_conjecture_4F5_literal`; $o(\log a)$ bound `eventually_calls_le`
Corollary 3	not stated separately; immediate from the statements for Corollary 2
Corollary 7 for intrinsic complexities	`uniform_intrinsic_bound_six`
Proposition 10	`AlmostAll.card_within_le`, `AlmostAll.card_D_le`, `AlmostAll.within_iff_D_le`, `AlmostAll.tendsto_card_within_loglog`
Theorem 11: recurrence and coefficient	`RegisterProgram.maxCost_step`, `RegisterProgram.Fixed.limsup_profile_le`
Theorem 11: program syntax	`RegisterProgram.Syntax.run_invokes`, `RegisterProgram.Syntax.accepts_eval`, `RegisterProgram.Syntax.eval_head_correct`, `RegisterProgram.Syntax.eval_head_exists`, `RegisterProgram.Syntax.arbitrarily_small_program`, `RegisterProgram.Syntax.eventually_eval_bound`
Operational correspondence (Section 7)	`RegisterProgram.Syntax.Machine.eval_iff_steps`, `RegisterProgram.Syntax.Machine.Step.context`, `RegisterProgram.Syntax.Machine.head_correct`, `RegisterProgram.Syntax.Machine.head_exists`, `RegisterProgram.Syntax.Machine.arbitrarily_small_program`

The refutation theorems state that no $r > 0$ satisfies $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b) \geq r \log_2 a$ for infinitely many $a \geq b \geq 1$, and that no $K > 0$ satisfies $2 \log_2 \max(x, y) \leq K \, \mathrm{calls}_{\mathrm{rem}}(\mathcal{A}, \gcd, x, y)$ for infinitely many pairs in $\mathbb{N}^2$, for either structure. The original $O(\log a / \log \log a)$ bounds combine the certificate bounds, which bound the intrinsic complexities by $G(a, b) + 2$, with `uniform_real_bound`; `eventually_calls_le` states the resulting $o(\log)$ bound.

A.2 The almost-all bound

Proposition 10 is checked in the module `ShortGCD.AlmostAll`, with the constant $40(2 + \ln N)$ in place of $16(2 + \ln N)$: the proof takes $B = a^{2^{-d}}/2$ separately for each pair instead of dyadic ranges. `AlmostAll.card_within_le` bounds the number of pairs $2 \leq b < a \leq N$ with a derivation of 1 of length at most d by $40(2 + \ln N)N^{2-2^{-d}}$; `AlmostAll.within_iff_D_le` identifies this with $D(a, b) \leq d$ on coprime pairs; and `AlmostAll.tendsto_card_within_loglog` proves the second claim in the form $D(a, b) > (1 - \varepsilon) \log_2 \log_2 N$ for all but $o(N^2)$ coprime pairs, for every $\varepsilon \in (0, 1]$.

A.3 Fixed programs and operational correspondence

Theorem 11 is checked in the eight modules `ShortGCD.RegisterProgram` through `ShortGCD.RegisterSmallStep`, which are included in the normal build and axiom audit. The theorem `RegisterProgram.maxCost_step` proves the fixed-parameter recurrence, and `RegisterProgram.Fixed.limsup_profile_le` proves its coefficient $3/p$. The syntax development gives finite body sets for the four internal symbols of Section 7. Generator runs translate into evaluation trees with exactly the same remainder count, including final verification. Every terminating evaluation is correct, an evaluation exists on every input, and the quantified bound holds; for a fixed p, the bound holds eventually with every coefficient greater than $3/p$.

In Lean we specialize the book's two-stack transition table to this language and program family. Closures represent substitution of actual values into terms; they are machine states, not additional program data. The control stack has its top at the head, and the value stack distinguishes individuals from booleans. Argument evaluation, zero testing, and branching are explicit transitions. The finite evaluation trees correspond to convergent machine computations, preserving both result and remainder count. For the forward direction, induct on the evaluation tree. Evaluate argument subtrees right to left, as prescribed by the machine's composition rule, then choose the indicated body at the internal call. Transition locality permits each subtree computation to be placed in its calling context [5, Table 1 and Lemma 2D.1, pp. 75–76]. A remainder node adds one external remainder call; a conditional evaluates its guard, makes one uncharged zero test, and evaluates only the selected branch. Variables and internal calls add no remainder calls. For the reverse direction, decompose a convergent computation at the completed argument evaluations and the selected body or branch, and induct on its length. This gives the same additive cost equations. Choices belong to individual call occurrences, and the language has no shared mutable state, so serializing the argument trees introduces no constraint on their choices. No successful tree uses a zero divisor or terminates through loop. The exact-cost equivalence is `RegisterProgram.Syntax.Machine.eval_iff_steps`, and transition locality is `RegisterProgram.Syntax.Machine.Step.context`. The machine versions of correctness and convergence follow, and `RegisterProgram.Syntax.Machine.arbitrarily_small_program` states the bound directly for finite transition sequences. This formalizes the machine needed here, not the book's theory for arbitrary structures and program vocabularies.

A.4 Integer parameters used in Lean

To avoid real rounding in the construction itself, the formal proof uses, for $j = \lfloor \log_2 n \rfloor \geq 32$,

$$p = \lfloor j/4 \rfloor, \quad k = 2^p, \quad h = 2^{2p} + 1, \quad J = \lfloor n/p \rfloor + 1, \quad \ell_1 = \lceil 3n/4 \rceil, \quad \ell_2 = \lceil n/3 \rceil.$$

Write $\widehat{T}(X) = \max_{a,b \leq X} G(a,b)$. The finite construction proves, for arbitrary natural parameters satisfying $k \geq 1$, $k^2 < h$, $Y \geq 1$, $2X \leq ZY$, and $X \leq Zk^J$,

$$\widehat{T}(X) \leq \max\{h + 1 + \widehat{T}(Y),\ h + 3J + 2 + 2\widehat{T}(Z)\}.$$

Taking $X = 2^n$, $Y = 2^{\ell_1}$, $Z = 2^{\ell_2}$ (so Y is the early threshold and Z the target modulus) gives the required hypotheses. Lean checks

$$(h+1)j \leq 64n, \quad (h + 3J + 2)j \leq 64n, \quad j - 2 \leq \log_2^{\mathrm{nat}} \ell_1, \log_2^{\mathrm{nat}} \ell_2, \quad 8\ell_1 \leq 7n, \quad 16\ell_2 \leq 7n.$$

The last estimates and $8j \leq 9(j-2)$ give a factor $63/64$ for either recursive contribution after clearing denominators. Taking $C \geq 4096$ and large enough for the finite initial segment proves $\widehat{T}(2^n)\lfloor \log_2 n \rfloor \leq Cn$ for all n. Comparing integer floors and ceilings with real base-two logarithms yields the stated real bound. These parameter choices change only absolute constants, not the proof mechanism or its asymptotic order. The refinement in `ShortGCD.SharpBound` uses the tunable parameters of Corollary 7, with recursive size bounds $2^{\lceil \alpha n \rceil}$ and $2^{\lceil \beta n \rceil}$. It applies the same finite recurrence, uses the original bound to justify a finite upper limit, and checks the rounding limits needed to preserve the coefficient 6.

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